

Catalin Niculescu

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INTRODUCTION

PhD student and Python developer with a strong interest in Data Science and Artificial Intelligence. Experienced in building data-driven applications, intelligent systems, and efficient software solutions that address real-world challenges. Curious and adaptable, I thrive in both independent and collaborative environments and continuously seek to expand my skills.

SKILLS

Technical Skills:

- Programming & Development – Python, Data Structures & Algorithms, OOP, REST APIs, Web Scraping, Git & GitHub, WordPress;
- Data Science & ML – Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-Learn;
- Databases – SQLite, MySQL, PostgreSQL;
- Embedded Systems & IoT – Raspberry Pi, Arduino, ESP32, IoT data collection;

Functional Skills:

- Critical Thinking & Problem Solving, Data Analysis & Interpretation, Cross-Functional Collaboration
- Process Automation & Optimization, Project Management & Teamwork

Languages – English, Romanian

WORK EXPERIENCE

IT Specialist – Data Science

The National Institute for Research & Development in Informatics - ICI Bucharest [07/2025 – Present]

- Analyze complex datasets using statistical methods, SQL queries, and Python-based data analysis tools.
- Produce data-driven reports and visualizations.
- Build predictive models and machine learning workflows for forecasting and pattern detection using scikit-learn.
- Design data preprocessing pipelines, including data cleaning, normalization, feature engineering, and missing-value handling.
- Contribute to scientific research and technical articles.
- Coordinate students during internship programs, providing guidance on technical tasks, project workflows, and applied research activities.

IT Specialist

SysCAD Solutions S.R.L. [07/2023 – 07/2025]

- Managing and maintaining IT infrastructure, including the RTK network and surveying equipment.
- Developing and optimizing software solutions for LiDAR and GPS data processing.
- Automating IT processes to improve workflow efficiency.
- Managing servers and databases for geospatial data storage and processing.
- Providing technical support, troubleshooting issues related to surveying equipment, RTK systems.

WORK EXPERIENCE

E-commerce Website Administrator & Developer – alphalookstore.com

AlphaLook Logistik S.R.L. [05/2024 – 11/2025]

- Developed and managed a full WordPress/WooCommerce e-commerce platform, including theme customization and plugin configuration.
- Wrote custom PHP scripts (functions.php), JavaScript enhancements, and CSS styling for UI/UX improvements and advanced WooCommerce functionality.
- Optimized SEO, site structure, metadata, and load performance (caching, image optimization, minification).
- Managed content, product catalog, and UX flows for an intuitive, conversion-oriented user experience.
- Implemented automation workflows: email notifications, order management, analytics, and marketing integrations.
- Ensured security and stability through Wordfence, regular audits, updates, and backups.

Electrical Engineer, Internship

SC Automobile Bavaria S.R.L. Baneasa [06/2022 – 09/2022]

- Assisted in diagnosing and troubleshooting electrical issues in vehicles.
- Conducted analysis and testing of electrical components using specialized equipment.
- Created technical reports and documentation for completed interventions.

Design Tehnician

S.C. Total Road S.A. [05/2020 – 12/2020]

- Designed and reviewed infrastructure plans for road construction projects.
- Utilized specialized software for technical drawings and simulations – AutoCAD.
- Collaborated with engineering teams to implement efficient design solutions.

EDUCATION

Doctoral Studies (Ph.D.)

Doctoral School of Electrical Engineering, Politehnica University of Bucharest [2025 – Present]

Thesis topic: "Autonomous environmental monitoring systems with artificial intelligence and distributed processing"

Master's Degree

Politehnica University of Bucharest [2023 – 2025]

Intelligent Instrumentation and Measurement Systems

Bachelor's Degree

Politehnica University of Bucharest [2019 – 2023]

Faculty of Electrical Engineering

COURSES & CERTIFICATIONS

Quantum Computing Road to Practitioner Program

IBM [02/2026 – 05/2026]

Verification: https://www.credly.com/badges/038b509b-4d91-49ab-8b34-48683d3fcb1f/linked_in_profile

Data Analysis with Python

IBM - Coursera [11/2025]

Verification: https://coursera.org/verify/0F07CMJU3FM

COURSES & CERTIFICATIONS

CS50's Introduction to Programming with Python (CS50P)

Harvard University - HarvardX – edX [08/2025 – 10/2025]
Verification:
https://cs50.harvard.edu/certificates/dea25fdb-5137-4205-8a76-9cec386d100c

CS50's Introduction to Databases with SQL (CS50SQL)

Harvard University - HarvardX – edX [07/2025 – 08/2025]

Certificate ID: 6bea7779d0f944baaa584b8f89e5f75b

Verification: https://courses.edx.org/certificates/6bea7779d0f944baaa584b8f89e5f75b

Junior Programmer Certified

Atelierele Ilbah [08/2023 – 11/2023]

Serial Number: S Nr. 0150080

Cambridge First Certificate in English(FCE)

Cambridge Assessment [2018]
Grade C
Certificate Number: 500/2705/0

European Computer Driving Licence Core

ECDL Core Certificate [2012]
Serial Number: RO091732

EUROPEAN FUNDED PROJECTS

1. FIT 2.0 European Digital Innovation Hub (EDIH) – Accelerating AI-Driven Digital Transformation in Romania's Center Region [05/2026 – 04/2029]

National Institute for Research & Development in Informatics – ICI Bucharest

- Project role: Process Improvement Specialist
- Work on: Python / AI / Data Science / VR / digital transformation etc.

2. CiTyInnoHub 2.0 European Digital Innovation Hub (EDIH) – Shaping the Future of AI-Driven Digital Transformation with SMEs and PSOs in South-East Romania in Sync with the European AI Ecosystem

National Institute for Research & Development in Informatics – ICI Bucharest

- Project role: Process Improvement Specialist
- Work on: Python / AI / Data Science / VR / digital transformation etc.

CONFERENCES & HACKATHONS

XR Creator Con 2026 – XRCC 2026 Hackathon

Berlin, Germany - June 30 - July 03, 2026

- ***Finalist / Community Voting Award Winner*** *- Developed and presented AgriSense AR, an augmented reality application integrating AI-based plant problem detection and an LLM-powered conversational assistant. The project was selected as a finalist and won the Community Voting side quest award.*

DISB 2026 - Digital Innovation Summit Bucharest

Bucharest, Romania - March 10-12, 2026

- *As a staff member at the booth of the National Institute for Research and Development in Informatics - ICI Bucharest, I presented and demonstrated virtual reality applications and robotic systems, contributing to the dissemination of research results and the promotion of technology transfer to the general public.*

14th International Symposium on Advanced Topics in Electrical Engineering (ATEE 2025)

Bucarest, Romania - October 09-11, 2025

- ***Presenter / Author** – Presented and published the paper "Real-Time Environmental Parameter Monitoring and Temperature Forecasting Using a Custom IoT Weather Station and Linear Regression"*

International Unmanned Aircraft Systems (UAS) Conference

Bucharest, Romania - September 25-26, 2024

- ***Speaker/Moderator** - presenting the geographical areas, limitations and protection of airports for flying with unmanned aircraft*

5th edition DevCon

Bucharest, Romania - November 01-02, 2023

- *Attending talks about new technologies, especially Artificial Intelligence*

PUBLICATIONS

1. ***C.-A. Niculescu, F. Argatu, G. Seritan, F. Adochiei, B. Enache,***
" Real-Time Environmental Parameter Monitoring and Temperature Forecasting Using a Custom IoT Weather Station and Linear Regression ",
Proceedings of the 14th International Symposium on Advanced Topics in Electrical Engineering (ATEE 2025), Bucharest, Romania, indexed in IEEE Xplore and Web of Science Core Collection, 2025,
DOI: 10.1109/ATEE66006.2025.11299969
Available at: https://ieeexplore.ieee.org/document/11299969
https://www.webofscience.com/wos/woscc/full-record/WOS:001741236900101
2. ***M.-V. Nicolae, I.-A. Mocanu, R.-A. Dutescu, C.-A. Niculescu,***
"A Structured Security and Privacy Assessment of Unity Authentication for Immersive Applications ",
Romanian Cyber Security Journal, vol. 8, nr. 1, pp. 13-34, 2026,
DOI: 10.54851/v8i1y202602
Available at: https://doi.org/10.54851/v8i1y202602

LANGUAGE SKILLS

Mother tongue(s): **Romanian**

Other language(s):

English	**French**
LISTENING C1 **READING** C1 **WRITING** C1	**LISTENING** A2 **READING** A2 **WRITING** A1
SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1	**SPOKEN PRODUCTION** A1 **SPOKEN INTERACTION** A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user